MyYangaX

MyYanga v5 with Lovable Virtual Wear UI

Document date: 5 August 2026

Prepared for ZonicMe Limited - Abuja, Lagos, Diaspora

Application overview, personas, technical architecture, SWOT, market analysis, expansion roadmap, screenshots, and deployment brief.

Table of Contents

1. Application Overview & Features
   - What is MyYangaX?
   - Core pillars
   - MyYangaX branding differentiators
   - Feature gates (monetization)
2. Capabilities by Persona
   - 2.1 Users & Reviewers
   - 2.2 Brands (Creators)
   - 2.3 Admin & Platform Operator
3. Technical Architecture
   - 3.1 Stack overview
   - 3.2 Frontend architecture
   - 3.3 Edge functions
   - 3.4 Virtual Wear pipeline
   - 3.5 Database migrations (001-011)
   - 3.6 Hostile audit (3-tier quality gate)
   - 3.7 Testing vs Production
4. SWOT Analysis
   - Context: African fashion tech, Nigeria market
5. Market Analysis
   - 5.1 Digital fashion & virtual try-on in Africa
   - 5.2 Addressable segments
   - 5.3 Competitive positioning
6. Digital Market Expansion
   - 6.1 Geographic growth scenarios
   - 6.2 Revenue streams
   - 6.3 Partner expansion (AdSpotX model analogue)
   - 6.4 Three-year expansion roadmap
7. Screenshots
   - Virtual Wear - Lovable UI
   - Discovery & engagement
8. Installation & Deployment Brief
   - Prerequisites
   - Quick start
   - Environment variables

MyYanga v5 with Lovable Virtual Wear UI Version: 5.0.0 (MyYangaX staging handoff) **Document date:** 5 August 2026 **Prepared for:** ZonicMe Limited - Abuja - Lagos - Diaspora **Confidential**

1. Application Overview & Features

What is MyYangaX?

MyYangaX is the branded staging handoff for **MyYanga v5** - a pre-production African fashion and lifestyle platform that connects fashion lovers, African creatives/brands, and the platform operator (ZonicMe). The August 2026 release integrates the **Lovable Virtual Wear UI**: a questionnaire-driven styling flow that curates up to four looks from a **paid brand pool**, then renders photorealistic or canvas try-on results.

> *Discover African creatives. Engage with their work. Patronize them directly. See yourself in their fashion before you buy - and let fashion reels find you on your schedule.*

Core pillars

Pillar	Description
Virtual Wear	Lovable-style stage machine: landing -> questionnaire -> loading -> results grid -> try-on. Paid brands only (is_promoted). Client-side curateVwLooks() - no separate curation edge.
Smart Search	Multi-parameter AI discovery: His/Hers, country, event, style, colour, piece, price band - every filter narrows together.
Reels	Member-controlled fashion reel delivery: in-app inbox, email, web push, WhatsApp - cadence and quiet hours in Settings.
Discover & Shop	Promoted brand pool with equal WhatsApp / Instagram / Virtual Wear / Shop actions. E-vetted brands unlock gateway-split checkout.
Editorial & Blog	Carousel programming + acquisition blog funnel with guest comments.
Admin & Governance	Full operator console: content, monetization, e-vetting (CAC), analytics, insights, reel config, access control.

MyYangaX branding differentiators

- **Lovable questionnaire flow** - selfie moderation, gender, event, vibes, budget, age group, modesty slider; profile pre-fill from member settings.
- **Paying brand pool** - only brands with creator_promotion / is_promoted rotate into Virtual Wear and Discover; fair no-repeat until pool cycles.
- **Try-on pipeline** - Replicate IDM-VTON -> Gemini image edit -> on-device canvas fallback; imagesDiffer() guard; garmentArt SVG when DB rows lack images.
- **Trust platform** - MyYanga is not merchant of record; Flutterwave/Paystack gateway split settles funds to designers.

Feature gates (monetization)

Gate	Feature
virtual_wear	Virtual try-on
smart_search	Smart Search AI

pyl_entry	Post Your Look competition entry
premium	Premium membership
events	Event access
creator_promotion	Paid placement in Discover + VW pool
retail_checkout	Marketplace bag (gateway split - not an entitlement gate)

Payment provider cascade: **Flutterwave -> Paystack -> Stripe**.

2. Capabilities by Persona

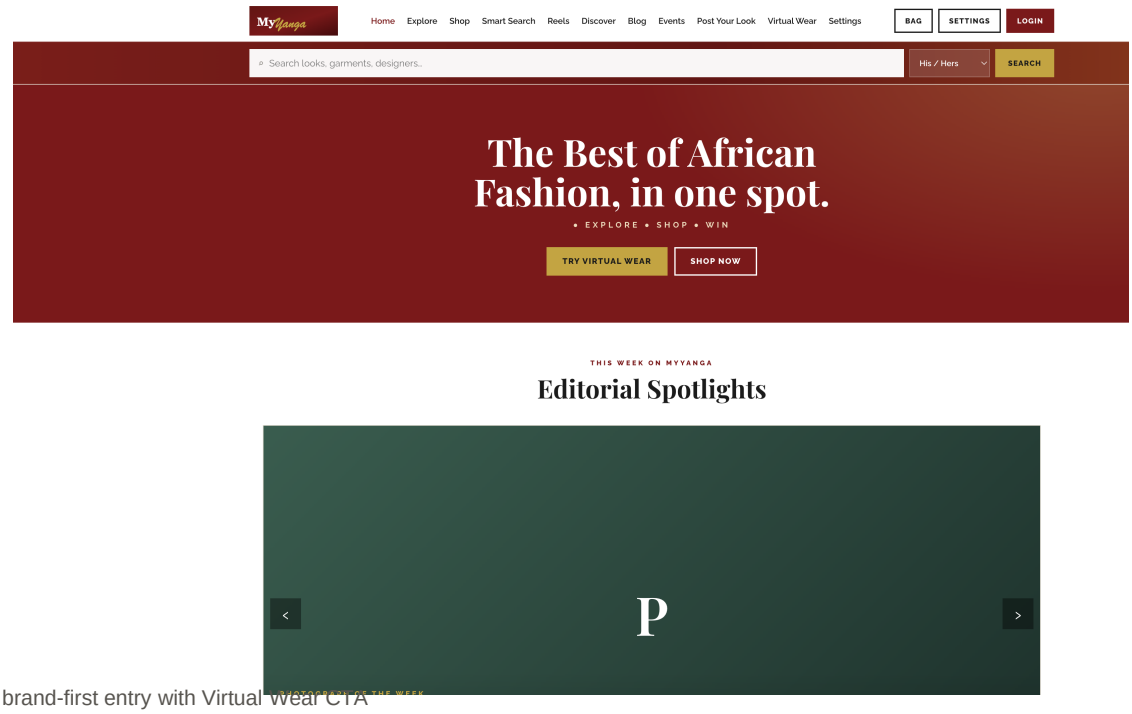
2.1 Users & Reviewers

Authentication (priority order)

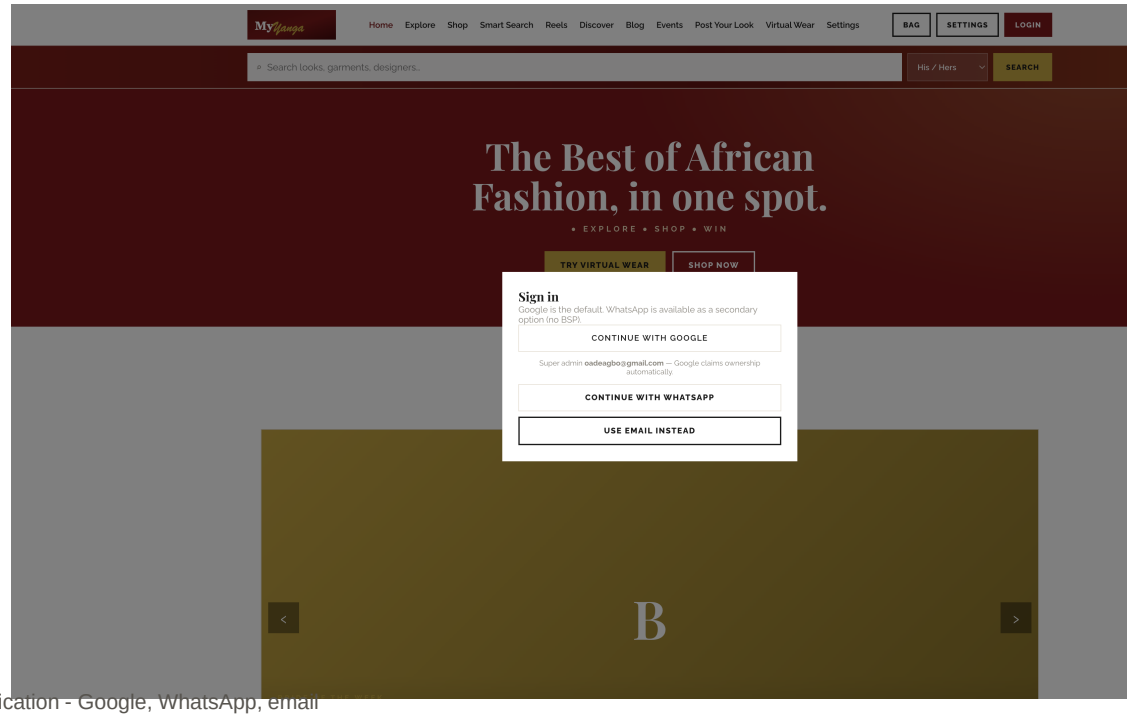
Method	Implementation
Google OAuth	Primary CTA; founding email oadeagbo@gmail.com auto-claims super admin
WhatsApp OTP	Secondary; Meta Cloud API via auth-whatsapp edge (no BSP)
Email magic link / password	Tertiary path via Supabase Auth

Member features

Area	Capabilities
Virtual Wear	Lovable landing -> questionnaire -> up to 4 curated looks -> Try on me -> share/download look
Smart Search	Filter garments and assembled look decks by gender, geography, occasion, style, colour, piece, price
Reels	In-app inbox; channel toggles; full schedule in Settings
Settings	Profile (name, email, phone, city, country, fashion preference, bio); reel cadence, days, timezone (Africa/Lagos default), quiet hours
Discover	Promoted brands; patronize via WhatsApp, Instagram, Virtual Wear, Shop
Shop	PDP, Bag, checkout for e-vetted brands only
Blog	Editorial acquisition funnel; share; guest comments
Competitions	Post Your Look; voting; shareable /look/{code} pages



Home - brand-first entry with Virtual Wear CTA



Authentication - Google, WhatsApp, email

2.2 Brands (Creators)

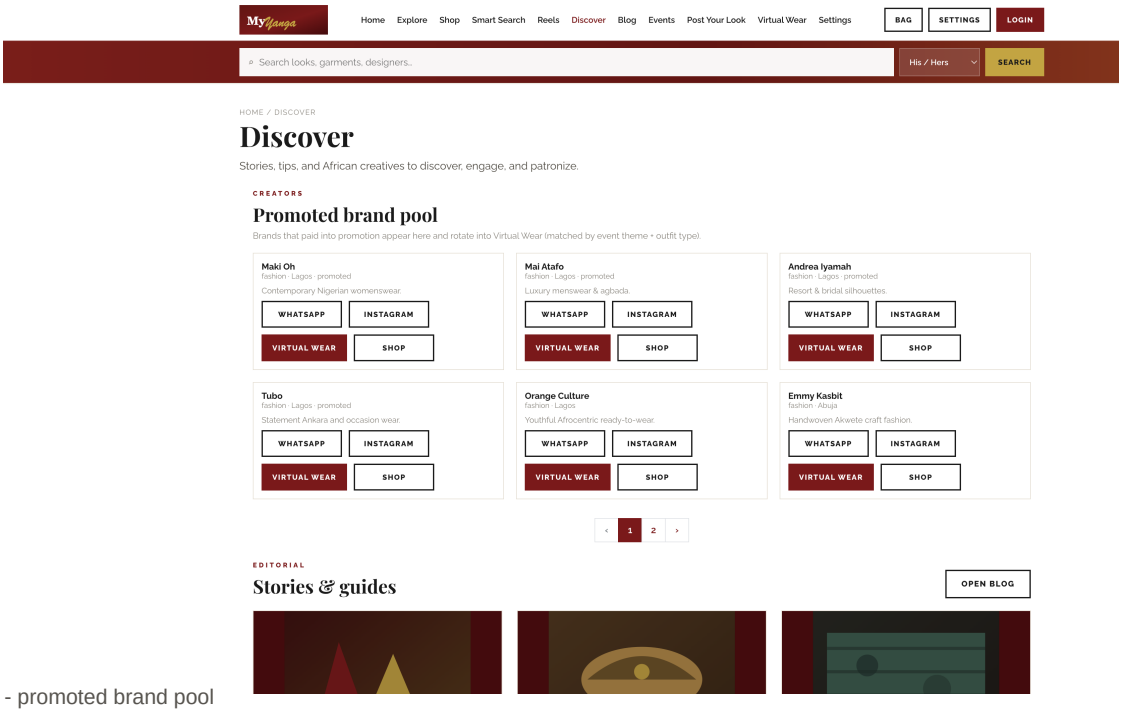
Brands are onboarded through **Admin E-vetting / CAC** and monetize via **promotion seats** and optional **e-vetted retail**.

Capability	Detail
Promotion seat	Pay creator_promotion -> is_promoted -> appears in Discover + Virtual Wear paid pool
Garment catalog	main_outfit and accessories with gender, event tags, style, price; matched by VW filters

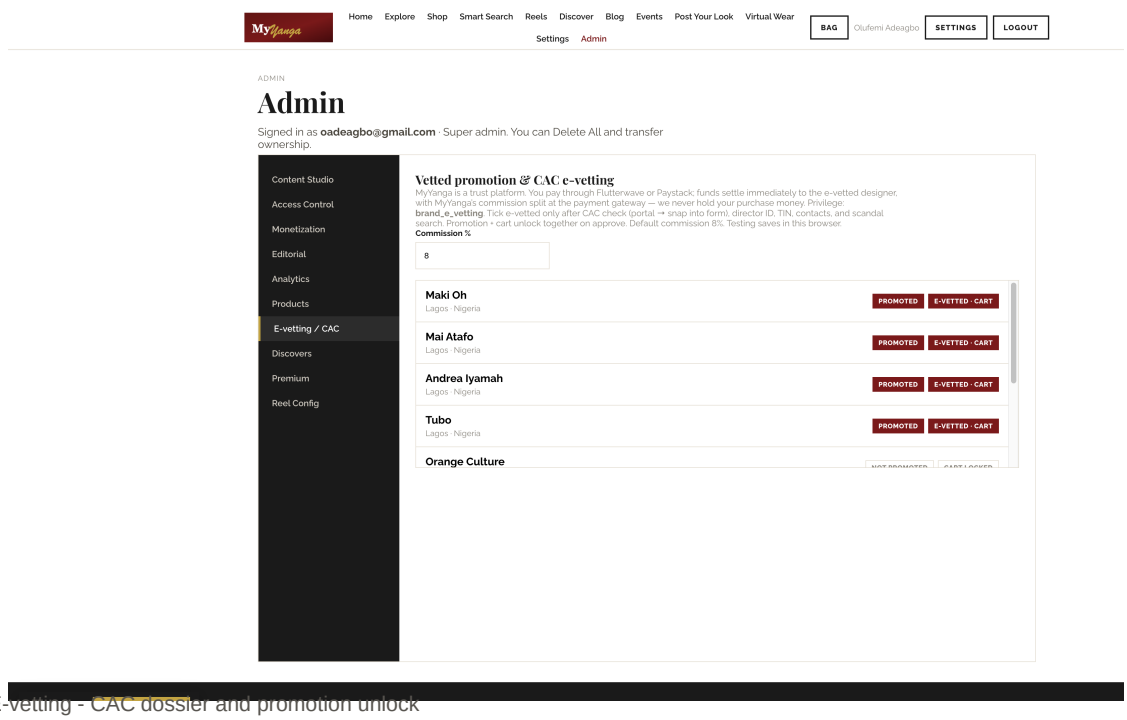
E-vetting (CAC)	Admin verifies RC, director ID, TIN, contacts, scandal search -> is_e_vetted unlocks Add to Bag + Flutterwave subaccount
Brand placements	Reel Config: brand ad slots per reel (max 0-15 slots)
Patronage channels	Equal-weight WhatsApp, Instagram, Virtual Wear, Shop on Discover cards
Analytics (Insights)	Brand-scoped insight packs: colours, VW renders, price bands, engagement; exportable via orbit_brand_key
Settlement	Gateway commission split at checkout - MyYanga never holds bag total

E-vetting workflow (Admin -> E-vetting / CAC)

- Open CAC portal / search brand RC
- Snap listing into form (legal name, RC, address, director, TIN)
- Complete contacts, socials, adverse-media notes
- Upload director ID shot
- Tick CAC verified + scandals cleared
- Approve e-vet + unlock cart -> sets is_e_vetted + is_promoted



Discover - promoted brand pool



Admin E-vetting - CAC dossier and promotion unlock

2.3 Admin & Platform Operator

Roles hierarchy

Role	Powers
Founding owner (super admin)	oadeagbo@gmail.com - auto-claimed on Google sign-in; sole delete_all; transfer super admin
Co-owners	platform_co_owners table; super-admin-equivalent via is_super_admin()
Delegated admins	admin_privileges grants per-studio access
Tester	tester@myyanga.test / Test2026! for QA audits

Privilege catalog (PRIVILEGE_CATALOG)

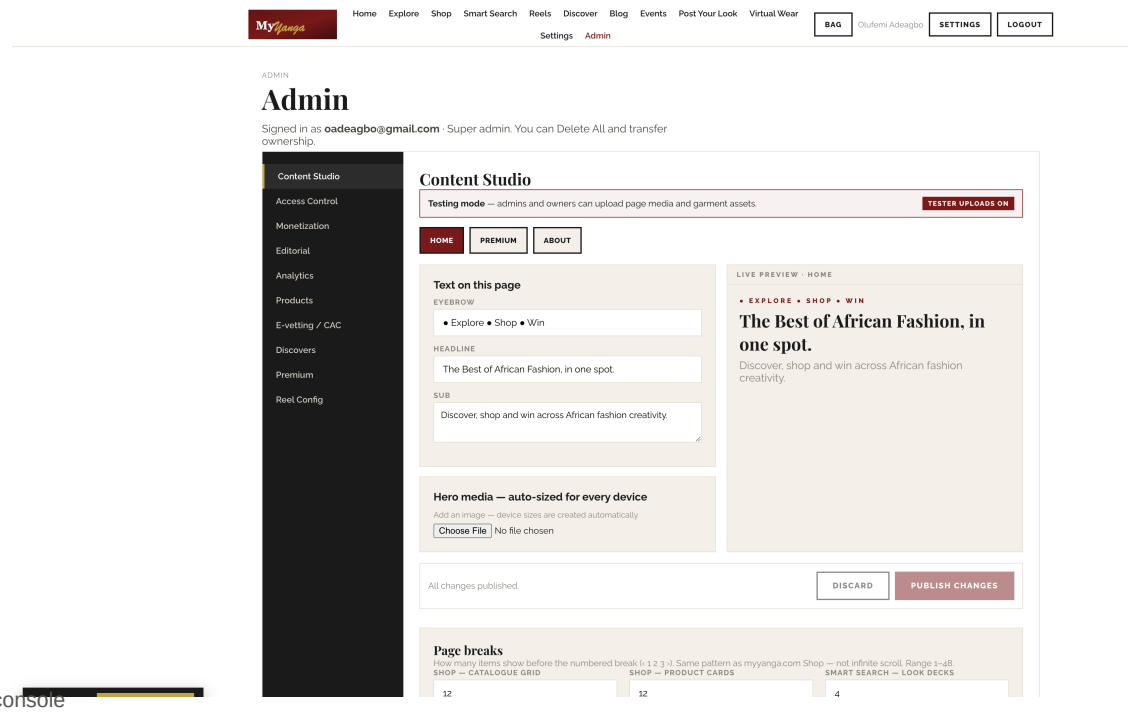
Privilege	Studio
content_studio	Page copy and media
editorial	Carousel programming
monetization	Feature prices and pay gates
analytics	Platform KPIs + fashion rollups
catalog	Products, brands, discovers
brand_e_vetting	CAC verify, promote, unlock cart
garments	Virtual Wear asset uploads
reel_config	Reel structure + brand placements
events	Event publishing
users_roles	Promote users to admin/brand

access_control	Directory view (mutations owner-gated)
delete_selected	Multi-select delete
delete_all	Owner-level wipe (dangerous)

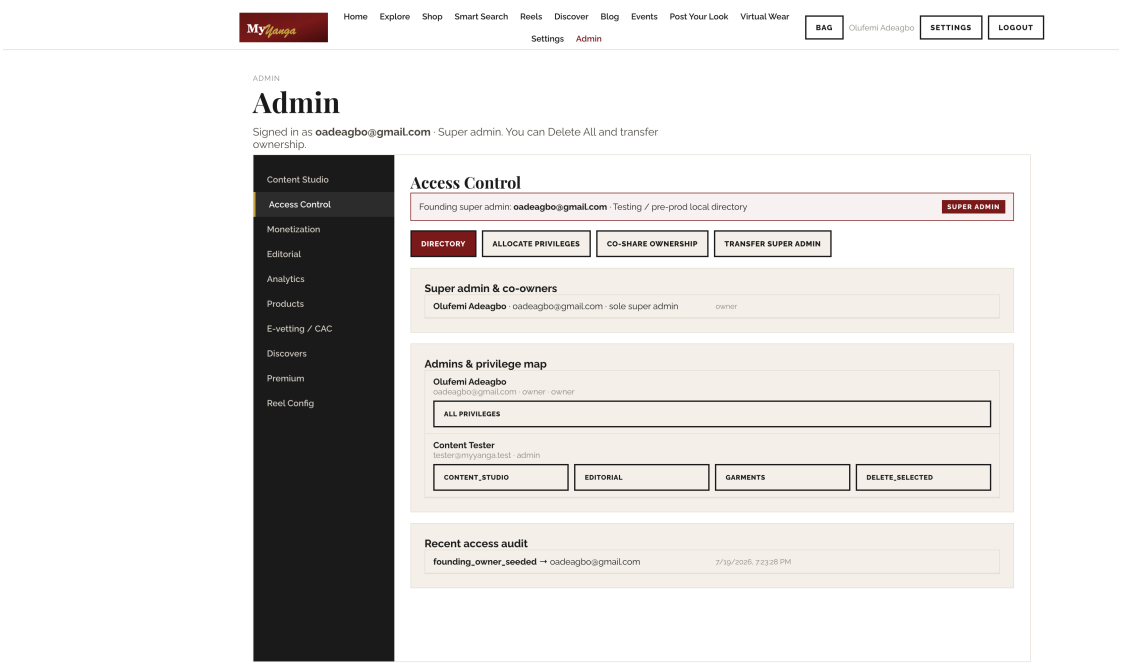
Admin studios

Studio	Purpose
Content Studio	Page copy/media, garment uploads
Access Control	Directory, privileges, co-owners, transfer, audit log
Monetization	Per-feature free/paid + provider order
Editorial	Outfit/Bride/Groom/Couple of the Week, creative focus
Analytics / Insights	Site KPIs, fashion rollups, AI assistant, brand packs
Catalog	Products, discovers, premium rows
E-vetting / CAC	Brand verification and promotion
Reel Config	Reel structure, brand ad slots, delivery metering

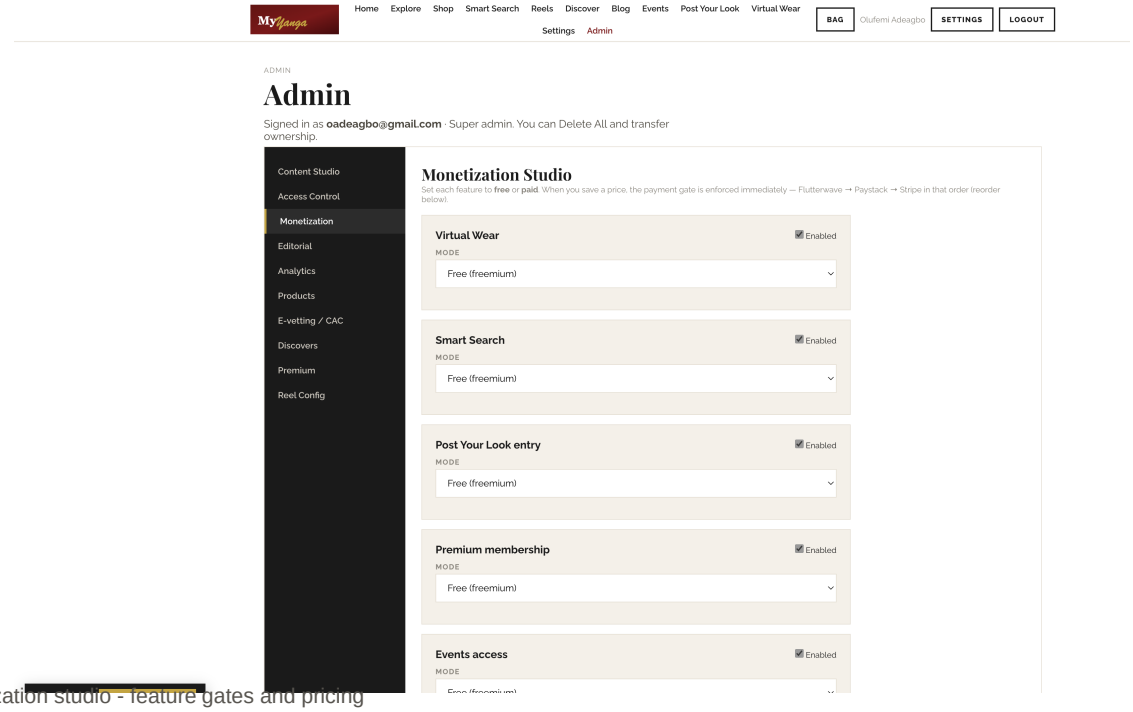
Security: rights mutations are **SECURITY DEFINER RPCs** with access_audit_log - UI is convenience, not the boundary. RLS on profiles prevents self-elevation.



Admin console



Access Control - super admin, co-owners, privilege grants



Monetization studio - feature gates and pricing

3. Technical Architecture

3.1 Stack overview



3.2 Frontend architecture

Module	Responsibility
src/myyanga-app.jsx	Shell, routing (view= query + /look/:code), design tokens, admin tabs
src/pages/VirtualWearPage.jsx	Lovable stage machine UI
src/lib/vwPool.js	curateVwLooks(), budget/vibe/modesty filters, no-repeat store
src/lib/virtualWear.js	aiTryOn(), compositeLook(), imagesDiffer(), renderLook()
src/lib/garmentArt.js	enrichGarmentImages() - SVG overlays for imageless garments
src/lib/selfieUpload.js	moderateSelfie(), prepareSelfieForTryOn()
src/lib/smartSearch.js	Client orchestration -> smart-search edge
src/hooks.js	useAuth() - Google / WhatsApp / email; founding owner claim
src/lib/dataBridge.js	Testing <-> Production mode switch

3.3 Edge functions

Function	Purpose	Auth
virtual-tryon	AI try-on cascade; get_feature_gate(virtual_wear) -> 402 if unpaid	JWT required
smart-search	Multi-parameter garment matching with array overlaps	JWT required
auth-whatsapp	WhatsApp OTP request/verify via Meta Cloud API	Public (rate-limited)
send-reel	Reel delivery engine (inbox, email, push, WhatsApp queue)	Service/cron
payments	Flutterwave/Paystack/Stripe feature gates + retail split	JWT + webhooks
ai-chat	Stylist / insights assistant (Anthropic Claude)	JWT

3.4 Virtual Wear pipeline

```
sequenceDiagram
    participant U as Member
    participant VW as VirtualWearPage
    participant Pool as vwPool.js
    participant Edge as virtual-tryon
    participant AI as Replicate/Gemini
    participant Canvas as Browser Canvas

    U->>VW: Start styling (questionnaire)
    VW->>Pool: curateVwLooks(filters)
    Pool-->>VW: Up to 4 paid-brand SKUs
    U->>VW: Pick look -> Try on me
    VW->>Edge: selfiePath + garments + scene
    alt AI available
        Edge->>AI: IDM-VTON or Gemini
        AI-->>Edge: Rendered image
    else AI unavailable / 502
        Edge-->>VW: fallback: true
        VW->>Canvas: compositeLook()
    end
    VW->>VW: imagesDiffer(input, output)
    VW-->>U: Result + share
```

Key guards

- **PayGate** - client wrapper + server 402 on unpaid virtual_wear
- **Paid pool only** - creators.is_promoted / vw_pool_active
- **No-repeat** - localStorage worn-outfit set per device (cap 200 IDs)
- **imagesDiffer** - rejects unchanged output (hostile audit hard rule)
- **garmentArt** - runtime SVG when seed garments lack image URLs

3.5 Database migrations (001-011)

Migration	Scope
001_schema	Core tables: profiles, creators, garments, settings
002_owner_and_delete	Owner role, delete RPCs
003_seed	Development seed data
004_production_core	Production-ready core
005_monetization	Feature gates, entitlements
006_seed_production	Production garment seed (imageless -> garmentArt)
007_access_control	Co-owners, privileges, audit log, founding owner
008_analytics_events	Anonymized fashion analytics
009_marketplace_commerce	Bag, orders, Flutterwave subaccounts
010_brand_e_vetting	CAC dossier columns
011_blog	Blog posts, guest comments

3.6 Hostile audit (3-tier quality gate)

Tier	Scope	Verdict rule
Tier 1 - BUILD	install, lint, build, verify:production, unit tests	Must pass

Tier 2 - MOCK	Dev server + Playwright/DOM: routes, VW canvas input, Smart Search filters, auth modal, admin	Must pass for SHIP
Tier 3 - LIVE	Real Supabase edges: 401 without auth, array filters, try-on honesty	PARTIAL if skipped

```
npm run audit:hostile              # all tiers
npm run audit:hostile:mock-only    # tier 1 + 2 (38/38 PASS after Lovable integration)
npm run audit:hostile:live         # requires cloud creds
```

Proven: build, garment art, canvas compositing, route wiring, PayGate, pool exports. **Not proven:** 100+ concurrent AI try-ons, burst Replicate queue, p95 edge timeouts.

3.7 Testing vs Production

	Testing (default)	Production
Data	Seeded mocks + localStorage	Supabase Postgres + RLS
Auth	Google/WhatsApp/email; founding = owner	DB roles; WhatsApp via edge
Virtual Wear	Canvas compositing	Gemini/Replicate + canvas fallback
Pay gates	All free	Live gates + payments
Analytics	Local device queue	analytics_events table

4. SWOT Analysis

Context: African fashion tech, Nigeria market

Nigeria is Africa's largest economy and a global hub for African fashion (Lagos Fashion Week, Nollywood styling, owambe culture). Smartphone penetration exceeds 45%; WhatsApp is the default commerce channel. Diaspora remittances and cultural pride drive demand for authentic African fashion abroad.

Strengths	End-to-end v5 platform shipped; Lovable VW UI differentiated; paid pool fairness model; WhatsApp-native auth; trust-platform checkout (no MoR); hostile audit 80/80 SHIP; multi-parameter Smart Search; member-controlled reels
Weaknesses	AI try-on latency (15-90s); no job queue for burst traffic; seeded garment art placeholders; blog CMS thin vs editorial studio; webhook signature depth; enterprise scale unproven
Opportunities	Nigeria -> Ghana/Kenya/SA expansion; UK/US/Canada diaspora; influencer commerce; newspaper/media embed (AdSpotX-style partner network for fashion editorials); telco data bundles; wedding/owambe seasonality
Threats	Instagram/WhatsApp informal commerce inertia; global fast-fashion apps; FX volatility affecting AI API costs; Replicate/Gemini rate limits; regulatory shifts on CAC/KYC; competitor virtual try-on from global players

5. Market Analysis

5.1 Digital fashion & virtual try-on in Africa

Trend	Africa-specific angle

Virtual try-on global CAGR ~25%	Diaspora sizing friction + trust deficit in DM commerce make "see yourself" especially valuable for Ankara, Agbada, Aso-Oke
Influencer commerce	Nollywood, Instagram stylists, owambe culture - reels on member schedule reduce fatigue vs spam
Creative economy policy	Nigeria Creative Industry Financing Initiative, AfCFTA - platforms that verify brands (CAC) align with policy
Mobile money / gateways	Flutterwave/Paystack dominance - gateway split model fits local payment rails

5.2 Addressable segments

- **B2C fashion lovers** - Nigeria (primary), Ghana, Kenya, South Africa + UK/US/Canada diaspora
- **B2B creatives/brands** - Designers, fabric makers, photographers, MUA seeking discovery + direct patronage
- **B2B2C partners** - Media houses, event sponsors, retailers embedding MyYanga discovery (AdSpotX analogue: partner embed without building fashion tech)

5.3 Competitive positioning

Alternative	MyYangaX gap closed
Instagram / WhatsApp DMs	E-vetting, VW pool fairness, member reel clock
Global marketplaces	African creative curation; diaspora sizing narrative
Pure social fashion apps	Monetization for brands; CAC trust layer
Traditional e-commerce	Trust platform - funds settle to designers at gateway

6. Digital Market Expansion

6.1 Geographic growth scenarios

Phase	Markets	Focus
Year 1	Nigeria (Lagos, Abuja, PH)	Paid VW pool brands, owambe/wedding season, WhatsApp reels
Year 2	West Africa (Ghana, Senegal, Cte d'Ivoire)	Local designer onboarding, Accra timezone defaults
Year 3	Pan-Africa + diaspora (UK, US, Canada)	Multi-currency checkout, diaspora sizing content, partner media embeds

6.2 Revenue streams

Stream	Year 1 target	Year 2 target	Year 3 target
Brand promotion seats (creator_promotion)	50 brands NGN 150k/yr	200 brands	500 brands
Virtual Wear pay gate (virtual_wear)	5k paid try-ons/mo	25k/mo	100k/mo
Smart Search (smart_search)	Freemium -> 2k subs	10k subs	40k subs
Retail commission (gateway split)	2% on NGN 50M GMV	NGN 300M GMV	NGN 1B GMV

Insights / Orbit brand packs	Pilot 10 brands	50 brands	200 brands
Reel brand placements	20 slots/mo	100 slots/mo	400 slots/mo

6.3 Partner expansion (AdSpotX model analogue)

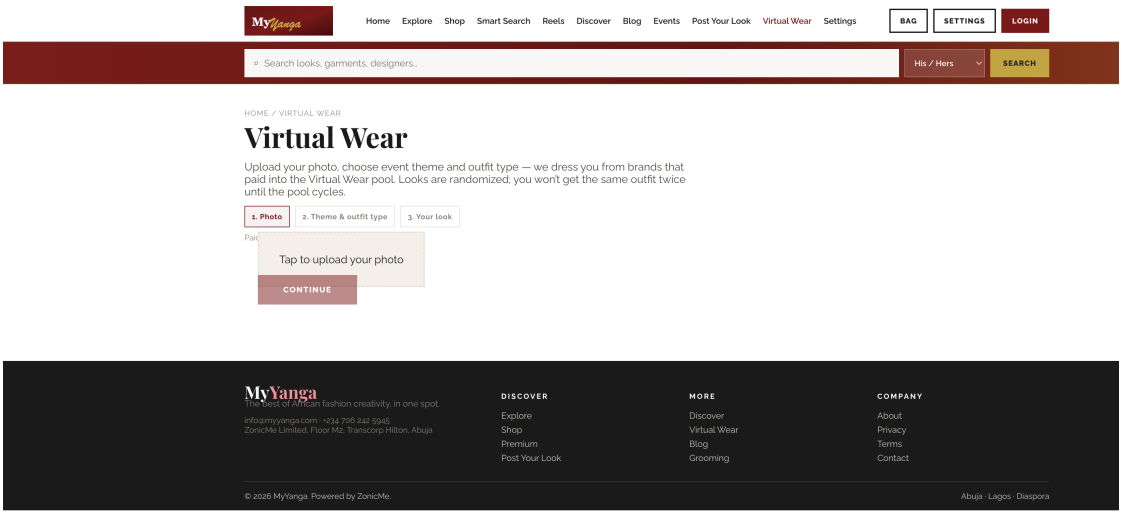
Partner type	Integration	Revenue share
Newspapers / media	Embed "Try on MyYanga" widget in fashion editorials	CPM + promotion seat referral
Retailers / malls	In-store QR -> VW for stocked designers	Commission on attributed sales
Event sponsors	LFW, owambe planners - gated event tickets + VW booths	Event gate + brand packages
Telcos	Zero-rated app access or bundle with data plans	Subsidy + promoted brand inventory

6.4 Three-year expansion roadmap

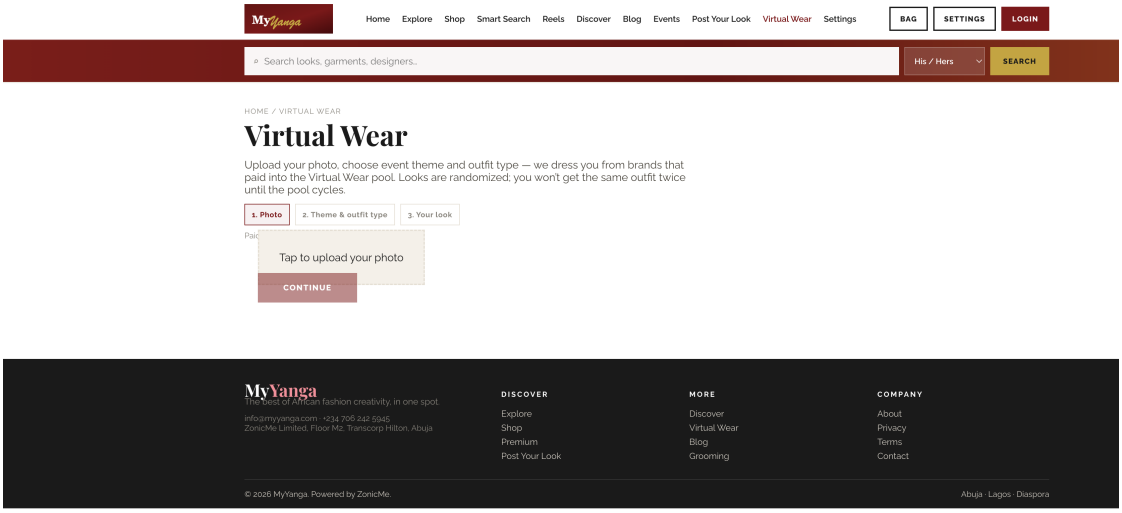
Quarter	Milestone	KPI
Q3 2026	MyYangaX staging -> production Amplify deploy	Hostile audit Tier 3 PASS
Q4 2026	25 paid promotion brands; LFW partnership pilot	10k MAU
Q1 2027	Ghana designer onboarding; reel WhatsApp Cloud live	50k MAU
Q2 2027	Job queue for AI try-on; rate limiting	p95 try-on < 45s
Q3 2027	Media partner embed (1 newspaper group)	100k MAU
Q4 2027	Diaspora campaign (UK/Nigeria corridor)	NGN 500M annualized GMV
2028	Kenya + SA launch; Orbit insights monetization	500k MAU
2029	Pan-African paid pool; 3 media partners	NGN 2B GMV; profitable unit economics

7. Screenshots

Virtual Wear - Lovable UI

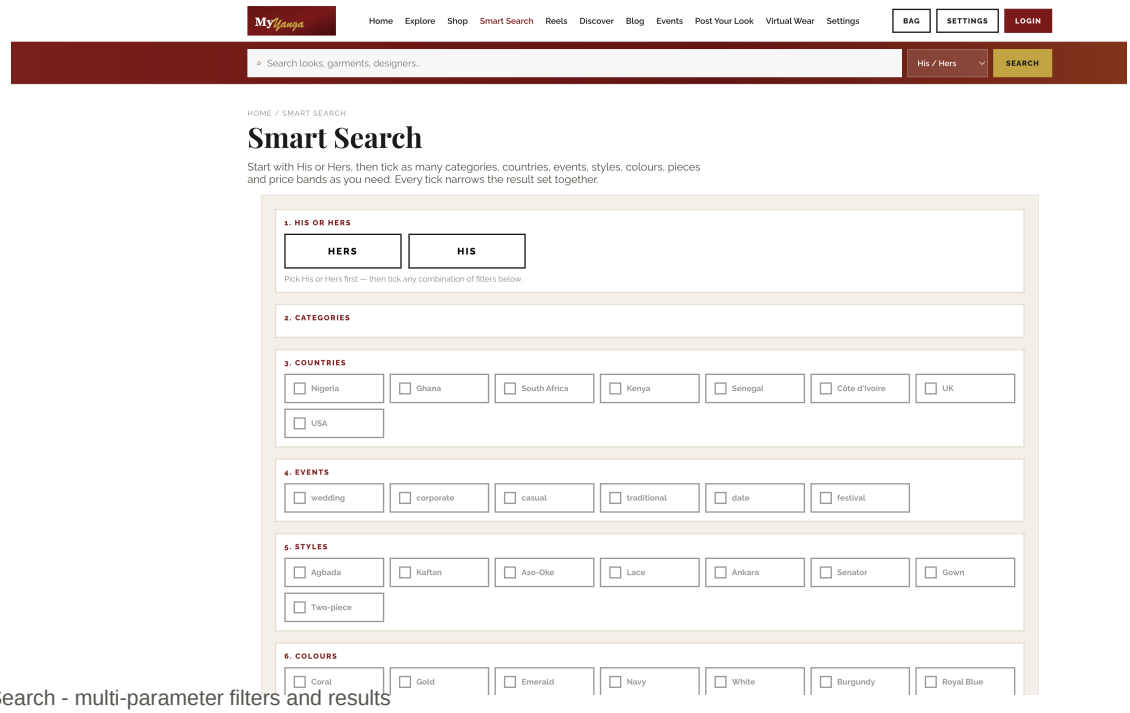


Virtual Wear landing - ankara pattern hero, Start styling CTA

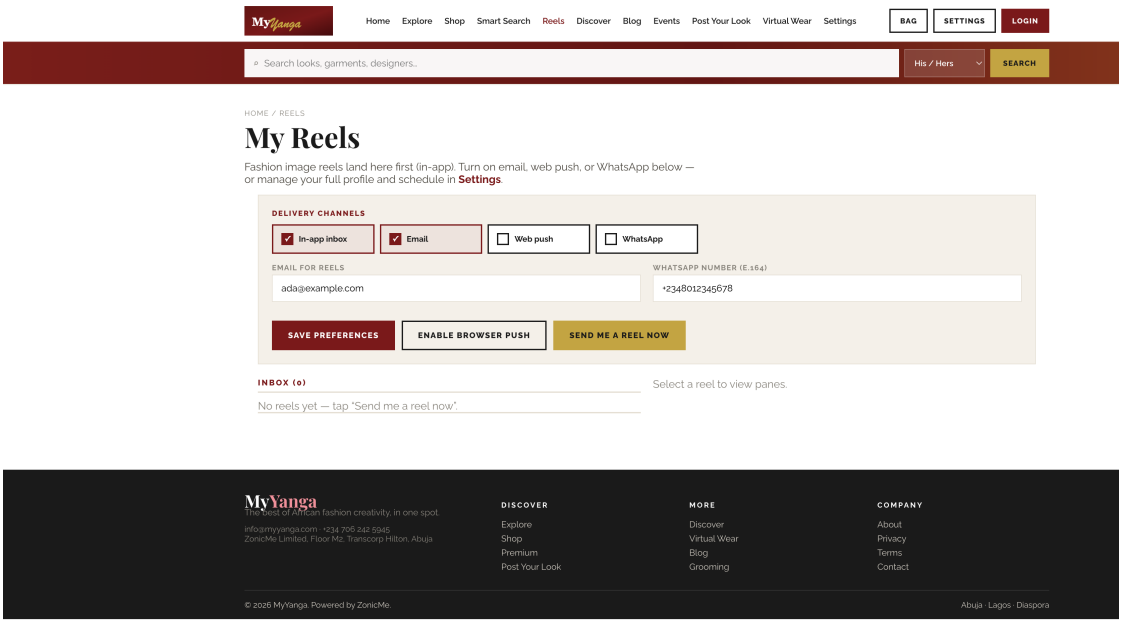


Virtual Wear questionnaire - selfie, gender, event, vibes, budget, modesty

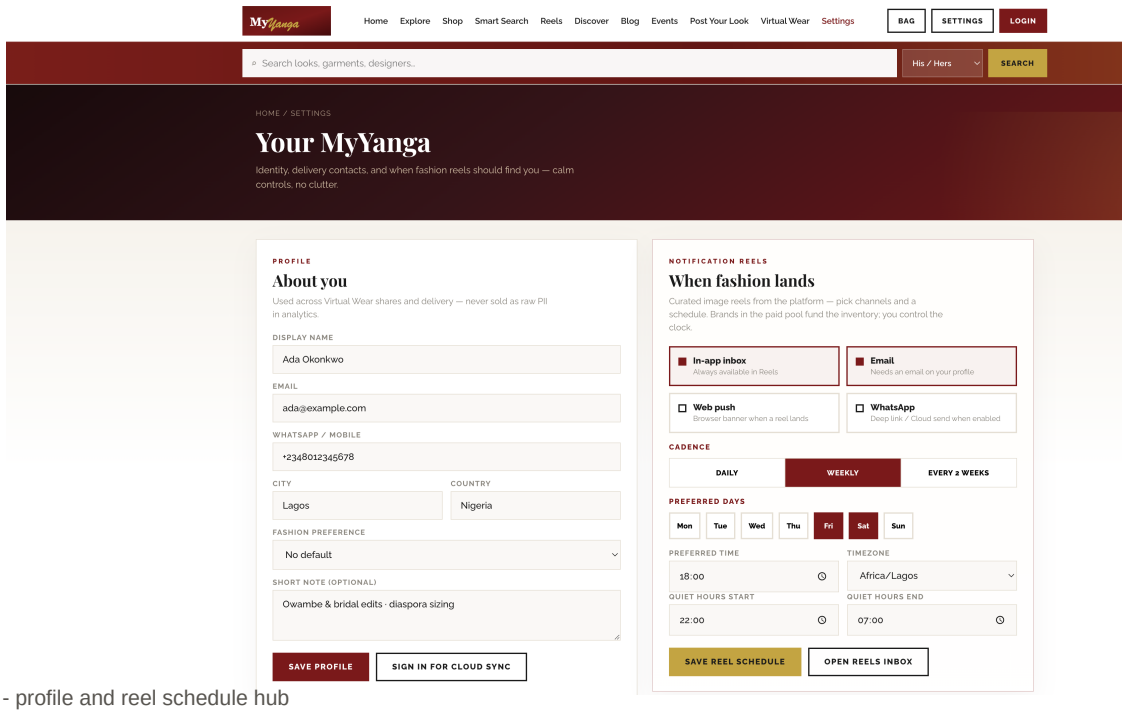
Discovery & engagement



Smart Search - multi-parameter filters and results



Reels inbox - member-controlled fashion delivery



Settings - profile and reel schedule hub

8. Installation & Deployment Brief

Source: docs/AWS_RESTAGE.md

Prerequisites

- Node.js 20+ (tested on 24.x)
- AWS Amplify (or S3 + CloudFront) for static hosting
- Supabase cloud project **or** local Supabase Tester for QA

Quick start

```
cd ~/Downloads/myyanga-v5   # or unpack MyYangaX.zip
npm install
npm run build
```

Environment variables

Secret	Where
VITE_SUPABASE_URL	Supabase -> Settings -> API
VITE_SUPABASE_ANON_KEY	Supabase -> Settings -> API
PUBLIC_APP_URL	Amplify domain (e.g. https://app.myyanga.com)
SUPABASE_SERVICE_ROLE_KEY	Server/edge only - never in client bundle

Edge function secrets: REPLICATE_API_TOKEN, GOOGLE_API_KEY, WHATSAPP_CLOUD_TOKEN, WHATSAPP_PHONE_NUMBER_ID, payment keys per docs/AWS_SECRETS_RUNBOOK.md.

Local QA without cloud Supabase

```
npm run supabase-tester:start    # :54322
npm run dev:tester               # Vite + .env.supabase-tester
USE_SUPABASE_TESTER=1 npm run audit:hostile
```

Deploy pipeline

```
supabase link --project-ref YOUR_REF
npm run sb:deploy                # all edge functions
supabase db push                 # migrations 001-011
npm run build                    # dist/
# Amplify: build command npm run build, output dist/, SPA fallback /* -> /index.html
```

Post-deploy smoke

```
LIVE_SITE_URL=https://your-amplify-url.amplifyapp.com \
VITE_SUPABASE_URL=... \
VITE_SUPABASE_ANON_KEY=... \
npm run audit:hostile:live
```

Hosting options

Platform	Role
AWS Amplify	Primary static SPA host; env vars at build time
Railway	Alternative Node/static host; same dist/ artifact
Supabase	Auth, Postgres, Storage, Edge Functions data plane

Rollback

- **Amplify:** redeploy previous build from console
- **Supabase:** restore DB snapshot or point env to previous project
- **Edge functions:** supabase functions deploy <name> from CLI history

Tester accounts

Email	Password	Role
tester@myyanga.test	Test2026!	Admin (tester)
oadeagbo@gmail.com	(Google OAuth)	Founding owner

Generate this PDF

```
npm run docs:myangax
# Output: docs/MyYangaX.pdf and ~/Downloads/MyYangaX.pdf
```

Prepared for ZonicMe Limited - MyYangaX complete platform overview. Confidential.